

A Comparative Evaluation of Probabilistic Electricity Price Forecasting Across Market Regimes

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ABSTRACT

Reliable probabilistic electricity price forecasting constitutes an essential prerequisite for optimal battery scheduling in deregulated energy markets. This study presents a comparative evaluation of probabilistic forecasting methods for the Dutch EPEX day-ahead market, examining Quantile Regression (QR), LASSO-regularized Quantile Regression with Bayesian Information Criterion selection (LQR-BIC), and Natural Gradient Boosting (NGBoost) against a naive benchmark. Our experimental framework employs an evaluation design spanning six distinct forecast origins from 2018 to 2024, encompassing stable conditions, the COVID-19 pandemic, and the 2022 energy crisis. Results reveal that linear quantile methods achieve superior average probabilistic accuracy, while NGBoost demonstrates the best calibration in half of the evaluated periods once adapted to prevailing market conditions. However, NGBoost exhibits vulnerability during rapid regime transitions, with degraded performance at the September 2021 crisis onset. These findings suggest context-dependent model selection for battery scheduling, where QR offers robustness across conditions while NGBoost provides superior uncertainty quantification during stable or established regimes.

KEYWORDS

Probabilistic forecasting, electricity prices, quantile regression, NGBoost, battery scheduling

1 Introduction

The accelerating integration of renewable energy sources into European electricity grids has fundamentally transformed wholesale market dynamics. Variable generation from wind and solar installations introduces substantial price uncertainty, creating both challenges and opportunities for flexible demand resources such as residential battery storage systems. Within this context, accurate probabilistic electricity price forecasting emerges as critical infrastructure for intelligent energy management (Weron, 2014; Nowotarski and Weron, 2018).

Point forecasts, while valuable for operational planning, provide insufficient information for risk-aware decision making. For battery scheduling under uncertain prices, quantifying forecast confidence is essential to balance expected arbitrage returns with potential losses. This necessitates probabilistic

forecasts that capture the full conditional distribution of future prices, enabling operators to implement strategies aligned with their risk tolerance (Nowotarski and Weron, 2018).

The Dutch EPEX day-ahead market presents particular forecasting challenges due to its substantial renewable penetration and interconnection with neighboring markets. Price distributions exhibit heavy tails, occasional negative values during excess renewable generation, and regime-dependent volatility patterns that intensified following the 2022 energy crisis. These characteristics demand forecasting approaches capable of capturing non-Gaussian distributional properties while maintaining computational tractability (Marcjasz et al., 2023).

This study contributes to the electricity price forecasting literature through a rigorous comparative evaluation of probabilistic methods under a multi-origin experimental design spanning diverse market conditions. Building upon Gielisse (2025), we extend prior work by implementing per-model feature selection strategies that respect distinct regularization properties, employing cross-validation to prevent data leakage (Cawley and Talbot, 2010), and evaluating performance across six forecast origins representing qualitatively different market regimes.

2 Methodology

Our dataset comprises hourly observations from the Dutch EPEX day-ahead market spanning January 2016 through August 2024, yielding 75,984 observations after removing 168 warm-up rows required for lag feature computation. The target variable is the day-ahead clearing price in EUR/MWh, while exogenous features include load forecasts and actual values from the ENTSO-E Transparency Platform, renewable generation forecasts for solar, onshore wind, and offshore wind capacity, and commodity prices for natural gas (TTF), carbon emissions (EUA), coal, and oil. Feature engineering produces 73 candidate predictors organized across several categories: temporal features capturing hourly, daily, and seasonal patterns, autoregressive features including price lags at 24, 48, 120, and 168 hours, rolling statistics computed over 24-hour and 168-hour windows, volatility measures, and interaction terms between renewable generation and temporal indicators. All rolling features are computed using strictly historical information to prevent look-ahead bias.

We evaluate three probabilistic forecasting methods against a naive benchmark, each employing distinct feature utilization strategies. The naive benchmark, following Gielisse (2025), generates prediction intervals using historical price volatility from the same hour and day-type. Quantile Regression (QR), introduced by Koenker and Bassett (1978), fits separate linear models for quantiles at levels 0.05, 0.10, 0.25, 0.50, 0.75, 0.90, and 0.95, with feature selection via correlation filtering to 30 predictors. LQR-BIC employs LASSO-regularized quantile regression with the regularization parameter selected by minimizing the Bayesian Information Criterion (Uniejewski and Weron, 2021), operating on all 73 features as L1 regularization performs implicit variable selection. NGBoost implements natural gradient boosting with a normal distribution assumption (Duan et al., 2020), using all 73 features with 5-fold cross-validation for hyperparameter tuning.

Our experimental protocol employs a multi-origin evaluation design to assess model robustness across market regimes. We define an anchor as the forecast origin date that separates historical training data from the out-of-sample test period. Six anchors span distinct market conditions. The September 2018 and December 2019 anchors represent stable pre-pandemic conditions, while March 2021 captures the pandemic recovery period. September 2021 marks the onset of the energy crisis, June 2022 corresponds to peak volatility, and March 2024 reflects post-crisis stabilization. For each anchor,

models are trained on 365 days of historical data and evaluated on a 30-day out-of-sample period. Target transformation employs a scaled inverse hyperbolic sine function fitted per-anchor to accommodate negative prices while stabilizing variance. Hyperparameter tuning uses 5-fold time-series cross-validation with rolling windows, incorporating purge gaps of 7 days and embargo gaps of 1 day to prevent information leakage between training and validation folds (de Prado, 2018). The NGBoost grid search evaluates 288 parameter combinations spanning estimator counts (100-800), learning rates (0.01-0.1), tree depths (2-5), and minibatch fractions (0.8-1.0), with Continuous Ranked Probability Score (CRPS) as the optimization objective.

Point forecast accuracy is assessed via Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE). Probabilistic forecast quality is evaluated using the CRPS, a proper scoring rule that measures the integrated squared difference between the predicted cumulative distribution function and the empirical distribution at the observed value (Gneiting and Raftery, 2007). CRPS generalizes MAE to probabilistic forecasts, reducing to MAE when the forecast is deterministic while rewarding both accuracy and calibration when the forecast is probabilistic, with lower values indicating better performance. Prediction Interval Coverage Probability (PICP) assesses calibration by computing the proportion of observations falling within a specified prediction interval (Khosravi et al., 2011). For a well-calibrated 90% prediction interval, empirical coverage should approximate the nominal 90% level. Substantial deviations indicate systematic over-confidence when coverage falls below 90% or excessive conservatism when coverage exceeds 90%.

3 Results

Table 1 presents performance metrics across all six anchors, revealing substantial regime-dependent variation. For point accuracy, LQR-BIC achieves the lowest mean MAE (13.09 EUR/MWh) and RMSE (17.92 EUR/MWh), followed by QR (MAE 13.28, RMSE 18.01). NGBoost exhibits higher errors (MAE 15.44, RMSE 20.84), primarily due to degradation at the September 2021 crisis onset. For probabilistic accuracy, QR achieves the best mean CRPS of 8.87 EUR/MWh. LQR-BIC follows with 9.33 EUR/MWh, while NGBoost achieves 10.82 EUR/MWh.

Calibration analysis reveals distinct characteristics. NGBoost achieves the best PICP in three of six anchors: September 2018 (92.9%), June 2022 (90.1%), and March 2024 (93.1%), demonstrating superior uncertainty quantification within familiar market conditions. QR maintains the most consistent coverage with mean PICP of 88.1%, approaching nominal 90% without severe failures. LQR-BIC shows higher variance, achieving excellent coverage in some periods (94.0% in March 2021) but catastrophic under-coverage at March 2024 (36.3%), likely due to overfitting to changed post-crisis volatility. The September 2021 anchor reveals critical regime transition differences. While QR maintains strong performance (CRPS 10.82, PICP 92.7%), NGBoost exhibits substantial degradation with CRPS of 20.95 and coverage collapsing to 66.5%. This vulnerability appears specific to rapid regime shifts where training data does not yet reflect new dynamics. At June 2022, after training on crisis-period data, all models perform comparably (CRPS 20.74-21.19), and NGBoost recovers to achieve the best coverage (90.1%).

4 Discussion and Implications

Our findings reveal complementary method strengths with direct implications for battery scheduling. For risk-neutral optimization, QR offers the most robust performance, achieving the best average CRPS with consistent calibration. For applications prioritizing well-calibrated intervals during stable

operations, NGBoost demonstrates superior uncertainty quantification, achieving closest-to-nominal coverage in half of evaluated anchors. LQR-BIC provides the best point accuracy but exhibits calibration instability limiting its utility for risk-constrained scheduling.

Practical implications are context-dependent. During stable periods, NGBoost’s superior calibration enables precise risk management with tighter scheduling margins. During market stress or regime changes, QR’s robustness provides more reliable risk bounds. The catastrophic coverage failures for NGBoost (September 2021) and LQR-BIC (March 2024) suggest operational systems should incorporate regime detection or ensemble approaches.

Table 1: Anchor-based model performance across market regimes

Anchor	Regime	Model	MAE	RMSE	CRPS	PICP 90%
Sep 2018	<i>Stable</i>	Naive	12.119	14.877	—	—
		QR	6.705	8.721	4.589	0.905
		LQR-BIC	6.149	8.248	4.204	0.894
		NGBoost	6.416	8.816	4.502	0.929
Dec 2019	<i>Stable</i>	Naive	11.621	13.903	—	—
		QR	4.561	6.429	3.212	0.834
		LQR-BIC	4.595	6.379	3.221	0.916
		NGBoost	4.687	6.525	3.314	0.838
Mar 2021	<i>Recovery</i>	Naive	10.517	14.782	—	—
		QR	8.453	12.669	5.645	0.876
		LQR-BIC	7.504	11.525	5.195	0.940
		NGBoost	7.997	11.576	5.631	0.855
Sep 2021	<i>Crisis</i>	Naive	36.518	43.929	—	—
		QR	16.408	21.601	10.817	0.927
		LQR-BIC	18.483	23.843	12.072	0.926
		NGBoost	29.861	37.545	20.956	0.665
Jun 2022	<i>Peak</i>	Naive	59.887	78.994	—	—
		QR	31.667	42.287	20.740	0.886
		LQR-BIC	30.123	41.500	20.875	0.784
		NGBoost	30.441	41.581	21.187	0.901
Mar 2024	<i>Post</i>	Naive	14.843	21.390	—	—
		QR	11.909	16.320	8.212	0.858
		LQR-BIC	11.671	16.022	10.392	0.363
		NGBoost	13.240	18.984	9.338	0.930

Several limitations merit discussion. Our specification employs a 365-day training window, 30-day test period, and specific hyperparameter grid representing one configuration among many. NGBoost may improve under alternative specifications: shorter training windows adapting more rapidly, different distributional assumptions beyond normality (Duan et al., 2020), or expanded search spaces. The normality assumption appears limiting given heavy-tailed crisis-period distributions (Marcjasz et al., 2023). Future work should investigate adaptive window selection, alternative distributions, and deep architectures including Temporal Fusion Transformers (Lim et al., 2021) and DeepAR (Salinas et al., 2020).

5 Conclusions

This study presents a multi-anchor evaluation of probabilistic electricity price forecasting methods for the Dutch EPEX market with application to household battery scheduling. Our experimental design spanning 2018-2024 reveals method-specific strengths. QR achieves superior average probabilistic accuracy and consistent calibration across conditions. NGBoost demonstrates the best uncertainty quantification during stable or established regimes but exhibits vulnerability during rapid transitions. LQR-BIC provides excellent point accuracy with calibration instability. These findings motivate context-dependent model selection and investigation of ensemble approaches for robust battery scheduling under diverse market conditions.

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